Introduction

Name Riju Bindua
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Education

2025–30*	**PhD**, Computer Science, McGill University
2021–25	**B.Tech.**, Mathematics and Computing (9.052/10), Indian Institute of Technology Delhi

Awards & Honors

2024	Mitacs Globalink Research Internship Award (GRI) by Mitacs
2023	Summer Undergraduate Research Award by Indian Institute of Technology Delhi
2022	Endowment Merit Scholarship by Indian Institute of Technology Delhi
2021	Female Topper in Rajasthan in JEE Main by National Testing Committee.
2021	Among Top 1% nationally in National Standard Examination in Chemistry (NSEC) by Tata Institute of Fundamental Research
2019	AIR 318, SA stream, Kishore Vaigyanik Protsahan Yojana (KVPY) by Indian Institute of Science (IISc)
2019	National Talent Search Examination (NTSE) scholar by National Council in Education Research and Training (NCERT)
2019	Regional Mathematics Olympiad (RMO) Tata Institute of Fundamental Research

Publications

1. Bindua, R., De, M., Garg, N. & Singla, K. *The Online Piercing Set Problem with Recourse* in *Proceedings of the Workshop in Approximation and Online Algorithms (WAOA)* (2025).

Preprint

2. Bindua, R., Hatami, H., Karimi, H. & Robere, R. *Lower Bounds for Approximate Sign Rank* Preprint. Available at https://arxiv.org/abs/2605.01038. 2026.
3. Bindua, R., Brüstle, T. & Scoccola, L. *Decomposing zero-dimensional persistent homology over rooted tree quivers* Preprint. Available at https://arxiv.org/abs/2411.19319v1. 2024.

Talks and Presentations

- A Hyperplane Avoidance Theorem for points in $\mathbb{R}^d$, *July 2026*, *Complexity Group Seminar*, McGill University, Montreal, Canada.

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- Tilting Theory *December 2025*, *BIRS*, Chennai Mathematical Institute, Chennai, India
- Assistive Navigation Technology for Visually Impaired *November 2024*, *Sixteenth Edition of Open House*, Indian Institute of Technology Delhi, India.
- Computing H_0 for D_4-filtered graphs *June 2024*, *Summer Research School: Applications of Representation Theory in Topological Data Analysis & Geometric Invariant Theory*, Université du Québec à Montréal, Canada.

Workshops Attended

- Czech Summer School on Discrete Mathematics (2026), University of West Bohemia, Pilsen, Czech Republic.
- ISM- Recent Advances in Probability and Combinatorics (2026), UQAM, Montreal, Canada.
- BIRS- Quivers in Representation Theory (2025), Chennai Mathematical Institute (CMI), online mode.
- AIS - Representation Theory of Quivers and BGG Category O for semi simple Lie Algebras (2024), IIT Kanpur, online mode. .
- Supervised by Prof. Naveen Garg and Prof. Amit Kumar in IIT Delhi CS Theory Winter Residency 2024.
- ISM- Summer Research School Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory, UQAM, Montréal, 2024

Teaching Assistant

- Tutoring Physics to a CEGEP student- Summer 2026 independently.
- COMP 360 (Algorithm Design)- Fall 2025 and Winter 2026 at McGill University.
- MTL 103 (Optimisation Methods and Applications)- Winter 2025 at IIT Delhi.

Thesis

Jan 2025–May 2025 **Online Geometric Hitting Set With Recourse**
Supervisor: Prof. Naveen Garg, Indian Institute of Technology Delhi
Co-supervisor: Prof. Minati De, Indian Institute of Technology Delhi

Often, relaxing the irrevocability condition in an online algorithm (recourse) improves the competitive ratio significantly. Any online algorithm computing the geometric hitting set for hypercubes in $\mathbb{R}^d$ has a lower bound $\Omega(d^2 logM)$ on the competitive ratio, where M is the ratio of side length of largest to smallest hypercube. In my thesis, we explored the improvements in the competitive ratio for this problem, on allowing recourse.

- Gave a 2-competitive algorithm with atmost 1 recourse per update for intervals in $\mathbb{R}$.
- Established that any algorithm giving optimum hitting set in $\mathbb{R}$ uses $\Omega(n)$ recourse per update where n is the number of intervals.
- Gave a $(1 + \frac{1}{k-1})$-competitive algorithm with atmost $2k$ recourse per update for intervals in $\mathbb{R}$, for any integer $k > 1$.
- Gave a 2^{d+1}-competitive algorithm with amortised 2 recourse per update for axis-aligned hypercubes in $\mathbb{R}^d$.

Research Internships

May 2024–July 2024 **Decomposing rooted tree modules with the elder rule**
Supervisor: Prof. Thomas Brüstle, Université de Sherbrooke

Often in Topological Data Analysis, only the zero dimensional homology H_0 of persistence modules is relevant. Computing H_0 takes cubic time for filtrations over any general quiver. However, for the linear case, using the so-called elder rule in TDA, it can be computed in log-linear time. In this project, we give a far-fetched extension to the elder-rule for rooted tree modules.

- Proved that for a graph filtration over rooted tree, there exist only finitely many indecomposable representations as its zeroth homology.
- Established the equivalence between these indecomposable representations and Kinser's reduced trees
- Gave a generalised inductive algorithm to get the indecomposable decomposition of the zeroth homology in quadratic time for rooted tree modules.

Other Experience

Assistech Lab, IIT Delhi

March 2023 - August 2023
Contributed to an assistive technology project aimed at helping visually impaired individuals locate and navigate building entrances independently.

- Developed a localization software system and trained it using a specialized dataset of entryway photos collected across campus.
- Developed a benchmarking tool to evaluate and test the performance of various assistive navigation applications.

SGBS Unnati Foundation

July 2021 - August 2021
Unnati is a non-government organization working to help individuals learn the skill set and get employed. I contributed to their digital marketing efforts.

- Researched digital marketing techniques and implemented off-page search engine optimization to spread awareness about the foundation's initiatives.
- Submitted the Unnati homepage link to over 20 web directories, improving online visibility and search engine rankings.